Does Manifold Steering Require Manifold Fitting?

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October 21, 2026

TL;DR: We reproduce the language-model experiments in [1] and test whether their results require nonlinear manifold fitting. First, we show that the reported isometry score is largely explained by the shared ordering of concept values imposed by the fitting method. Nevertheless, a stricter comparison of corresponding adjacent distances finds significant local alignment. Second, replacing the fitted splines with piecewise-linear paths between adjacent centroids reproduces the steering results, and removing the PCA projection does not reduce performance. Finally, we test whether the intervention generalizes when the concept value is separated from the output token using a multiple-choice task. The intervention only retains a moderate effect in this setting.

Manifold Steering Reveals the Shared Geometry of Neural Network Representation and Behavior

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https://github.com/goodfire-ai/causalab/tree/manifold_steering

1 Testing the claimed isometry

1.1 Construction of the isometry score

Background. This paper from Goodfire studies the relation between the representations of some concepts in the activation and behavior spaces. More precisely, they study cyclic (days/months) and sequential (letters/ages) concepts, for which the representations are known to have particular nonlinear geometry [2]. These geometries are now well understood as emerging from specific language statistics and symmetries [3].

Their goal is to study these representations in both the activation and behavior space and investigate the geometric relations between the two representations in each space. They claim (see below) to build an isometry between the two spaces and validate it using manifold steering.

Claim [1]

Uncovering isometric geometries in neural network representation and behavior. We use tasks where models output a distribution over a set of concepts with known structure. In each task, we fit an activation manifold $\mathcal{M}_h$ to internal representations and a behavior manifold $\mathcal{M}_y$ to model outputs (probability distributions over task-relevant concepts). We show the two geometries are tightly interlinked via a scaled isometry relation: geodesic distances on $\mathcal{M}_h$ align closely with those on $\mathcal{M}_y$, and neither match Euclidean distances.

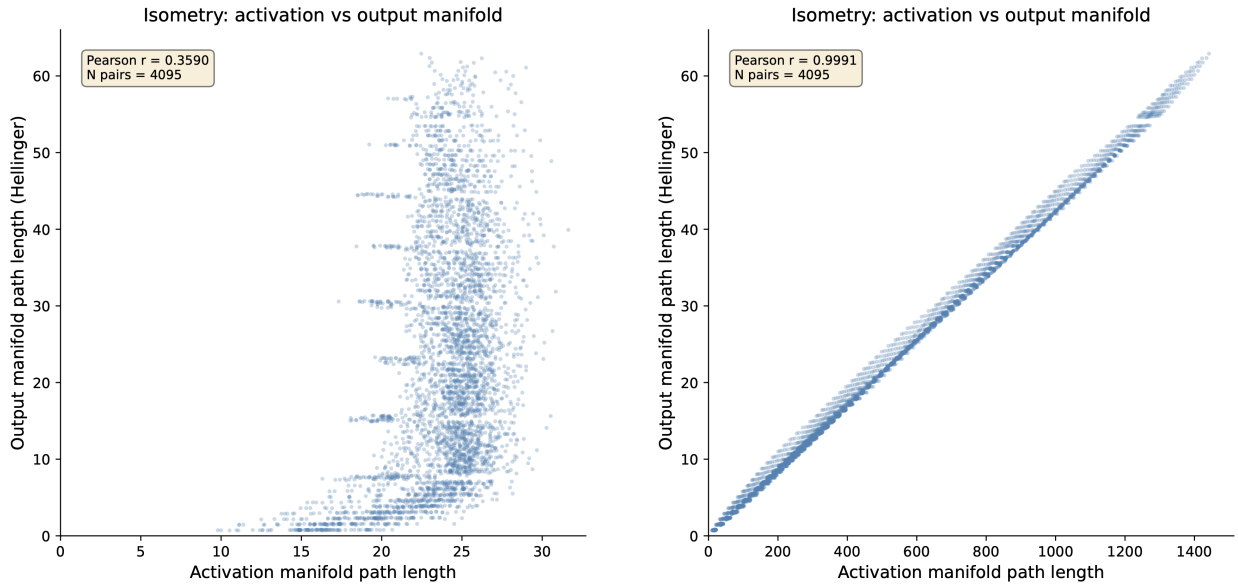


Figure 1: Reproduction of "scaled isometry" results for the ages category.

Details on the paper's construction. The paper focuses on representations built in the activation and behavior spaces, which are defined as:

- Activation space: $\mathbb{R}^n$ (outputs of layer 28 of Llama-3.1-8B)
- Behavior space: $\Delta^{|\mathcal{Z}|}$ (output probability distribution over tokens)

The concept distributions are extracted from the model's activations and output distribution for every prompt. Prompts are designed to elicit a specific concept (e.g. "What is k days after ...?") and centroids are computed by grouping prompts that share the same correct answer.

The paper details a method to fit a manifold in both spaces:

Activation space	Behavior simplex
1. PCA to 64 dimensions 2. Compute centroids 3. Fit cubic splines through centroids	1. Transform to Hellinger space using $p \mapsto \sqrt{p}$ 2. Compute centroids 3. Fit cubic splines through centroids

Importantly, the spline fit is conditioned on an ordering of the concept values. For sequential concepts, this ordering is given by the ground-truth ordinal index. For cyclic concepts, it is obtained using the angular position of the activation centroids in the first two principal components. The fitted paths therefore encode the concept ordering before the isometry score is computed.

1.2 The isometry score is dominated by concept ordering

The isometry score is built by computing the Pearson correlation between the upper-triangular entries of the pairwise distance matrices. Distances between two points are defined as the spline arc-length between them, and computed using integration along the spline. Because the spline is fitted to go through each value in order, distances between points can be decomposed as the sum of adjacent gaps between these two points:

$$d_{\mathcal{M}}(c_i, c_j) = \sum_{k=i}^{j-1} d_{\mathcal{M}}(c_k, c_{k+1})$$

Because the two splines use the same ordering, the distances associated with a given pair (i, j) contain the same number of gaps in both spaces. When the gaps have low dispersion, both distances are therefore determined mainly by $|i-j|$. This produces a high all-pairs correlation even when the lengths of corresponding gaps are unrelated. The score consequently conflates agreement in concept ordering with agreement in metric structure.

In practice, we observe that the manifold distance between two values is heavily correlated with the distance between the two values in the ordering[1]:

Distance correlation	Weekdays	Months	Letters	Ages
$\mathrm{corr}(d_{\mathcal{M}_h}(c_i, c_j), \lvert i-j \rvert)$	0.970	0.884	0.997	0.998
$\mathrm{corr}\big(d_{\mathcal{M}_y}(c_i, c_j), \lvert i-j \rvert\big)$	0.994	0.992	0.998	0.999

This strong dependence on concept ordering makes the isometry score difficult to interpret. The high values reported are directly a consequence of the correlation of manifold distances with the ordering of points, which itself is a consequence of the fitting method. We next test whether similarly high scores arise when the two paths share an ordering but no meaningful geometric correspondence.

1.3 Null models reproduce high isometry scores

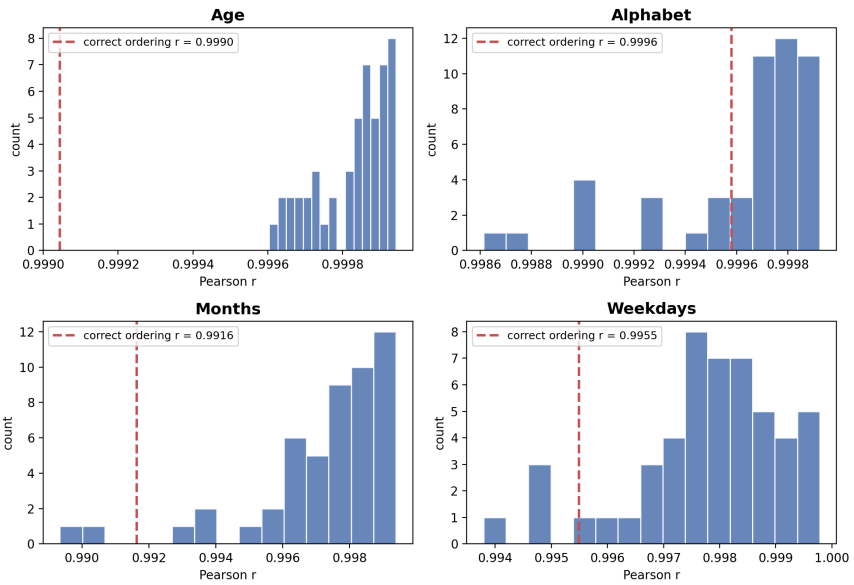
To further illustrate this point, we perform two additional experiments:

1. We build the manifolds using a random ordering of the concept values (same for both activation and behavior spaces) and report the same correlation results.

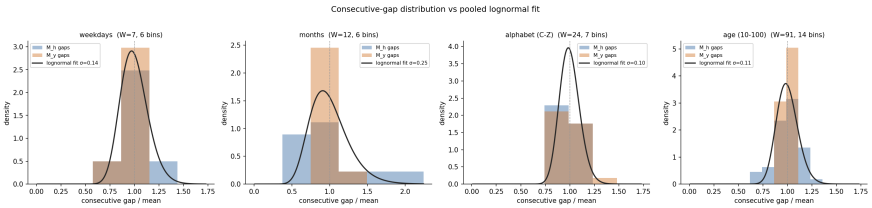
The isometry score remains extremely high after the concept values are randomly reordered. The score therefore rewards the shared ordering used to fit the two paths, even when that ordering has no semantic meaning. This shows that a high all-pairs correlation is not sufficient to identify a geometric correspondence beyond the shared ordering used to fit the two paths.

2. We show that the values obtained are simply a consequence of the low gap dispersion in both spaces.

[1]For cyclic concepts, the distance in the ordering is measured as $\min\{|i-j|, W-|i-j|\}$.



We fit the distribution of gaps (distance between two consecutive values[2]) to a log-normal distribution and then generate sequences of gaps from this fitted distribution. We then compute the correlation metric from the paper based on off-diagonal pairs of points to obtain a distribution of correlation values. The results show that the correlation values obtained in the paper fall within the expected range.



Domain	Pooled gap σ	Null all-pairs r: mean [5%, 95%]	Observed all-pairs r
Weekdays	0.14	0.963 [0.913, 0.991]	0.967
Months	0.25	0.950 [0.889, 0.985]	0.922
Alphabet	0.10	0.999 [0.9975, 0.9996]	0.9996
Age	0.11	1.000 [0.9993, 0.9999]	0.9990

Conclusion

The reported isometry score cannot distinguish a shared geometry between the two spaces from the simpler case of two ordered paths with low-variance gaps. The score therefore does not provide independent evidence for the claimed isometry.

[2]Distance can be either the Euclidean distance or the spline distance between two consecutive values. Plots are shown below for spline distance but results are similar using Euclidean distance.

1.4 Do corresponding local distances align?

Given that an ordering of concept values is implicitly assumed when fitting a manifold, a more natural question is:

Are the distances between consecutive points in the two representations correlated?

In practice, the distances between consecutive points exhibit some correlation, as detailed in Table 1:

Table 1: Correlation between corresponding adjacent gaps.

Concept	Gaps	Observed r	Null mean $\pm$ SD	Null 95th percentile	p
Weekdays	7	0.310	0.00 ± 0.41	0.60	0.24
Months	12	0.766	0.00 ± 0.30	0.49	0.0016
Alphabet	23	0.812	0.00 ± 0.21	0.36	$< 10^{-4}$
Age	90	0.519	0.00 ± 0.11	0.18	$< 10^{-4}$

The observed correlations are significantly larger than the permutation null[3] for months, alphabet, and age. The null distribution for weekdays is wider simply because of the smaller number of values.

These results show that the activation and behavior representations share more than the ordering of concept values. For three of the four concepts, a relatively long gap in activation space also tends to be relatively long in behavior space. However, this result does not establish a scaled isometry: correlation between corresponding gaps does not imply that their lengths are preserved up to a common scale factor.

2 Piecewise-linear paths reproduce manifold steering

Given that the isometry result is not as strong as the reported numbers suggest, we question whether the manifold construction built in the paper remains relevant.

A simple ablation is to **replace the fitted spline by linear interpolation between each value**, which builds a piecewise linear path across the concept. Since linear steering between two concepts usually works directly on activations without any prior PCA projection, we also question whether the PCA projection is necessary.

2.1 Spline and Euclidean gap statistics are similar

We first reproduce the experiment described in Section 1.4 and compare Euclidean distances with the spline distances inherited from manifold fitting.

For these two statistics, spline arc lengths and Euclidean distances give similar results, which suggests that the manifolds do not carry much additional structure beyond the piecewise linear interpolation.

[3]Let $a_i = d_{\mathcal{M}_h}(c_i, c_{i+1})$ and $b_i = d_{\mathcal{M}_y}(c_i, c_{i+1})$ denote the activation- and behavior-space gaps. We randomly permute the correspondence between the behavior-space and activation-space gaps. For each of 10,000 permutations π, we compute $r_\pi = \mathrm{corr}(a_i, b_{\pi(i)})$. This preserves both gap distributions while removing their position-wise correspondence.

Metric	Distance	Letters ($W = 24$)	Ages ($W = 91$)
Gap correlation	Spline-arc-length	0.812	0.519
	Euclidean	0.800	0.450
Gap CV (activation)	Spline-arc-length	11.2%	13.8%
	Euclidean	12.2%	14.2%
Gap CV (behavior)	Spline-arc-length	9.6%	3.7%
	Euclidean	10.0%	3.7%

2.2 Spline fitting does not improve causal interventions

In practice, the relevance of the concept geometry is validated by performing causal interventions in the activation space and observing the downstream effects in the behavior space. We reproduce Figure 4 from [1] and incorporate the two ablations mentioned above: piecewise linear steering paths with and without preliminary PCA.

For a given prompt, we steer the answer from the original prompt to a different answer, following the constructed path. We expect geometry-aware interventions to show smooth transitions between intermediate answers. Figure 2 compares manifold steering, piecewise linear steering with PCA, piecewise linear steering without PCA, and finally direct linear interpolation.

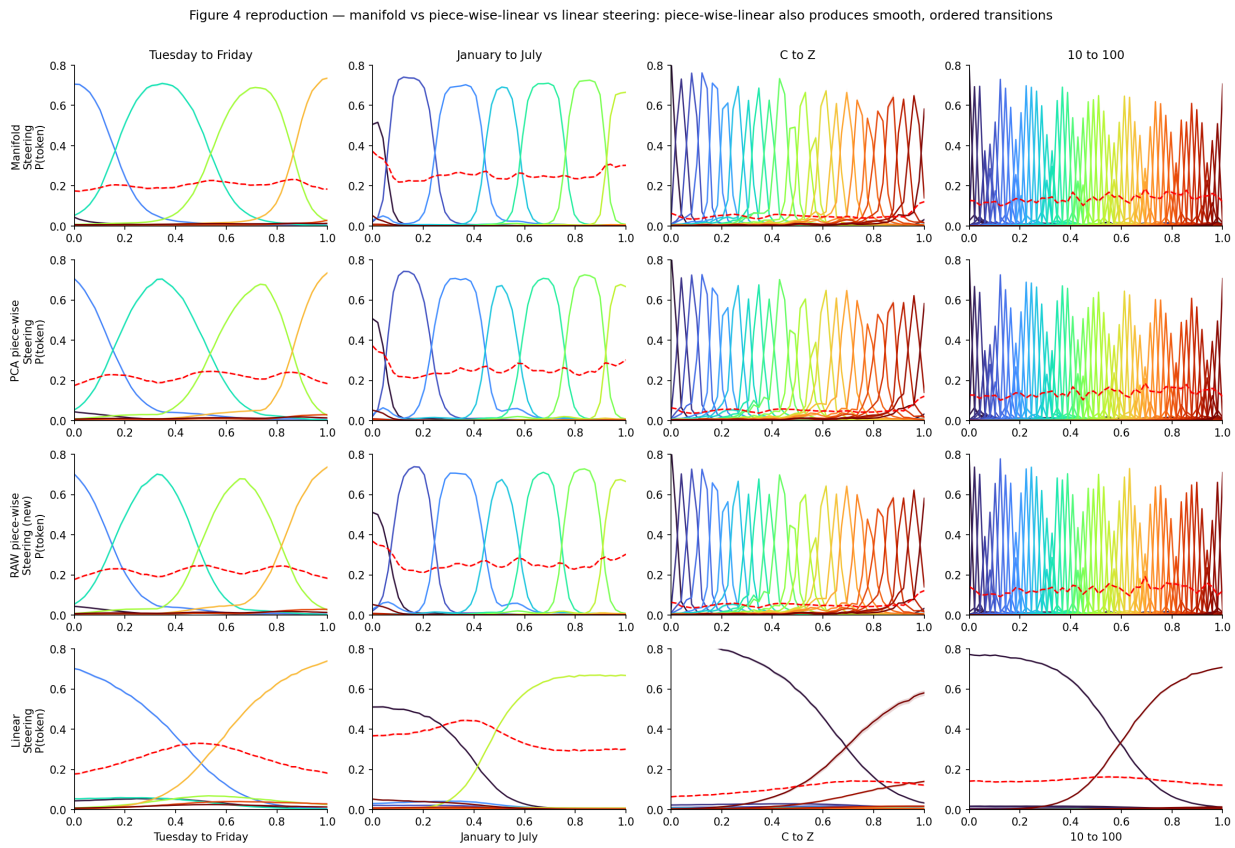


Figure 2: Reproduction of Figure 4 from [1]. Rows: manifold steering, piecewise linear steering with PCA, piecewise linear steering without PCA, and direct linear interpolation.

We observe that the causal interventions obtained by interpolating linearly between consecutive

values perform as well as the manifold interventions, which suggests that the complex manifold fitting procedure is not necessary to obtain these causal results. In particular, both ablations exhibit smooth transitions between consecutive values, as in the manifold case.

Conclusion

Nonlinear manifold fitting is not necessary to reproduce the reported steering results. Piecewise-linear steering between consecutive centroids yields near-identical intermediate output distributions. The PCA projection also provides no visible improvement.

For completeness, we perform similar ablations on multi-dimensional concepts in Appendix A.1.

3 Does the intervention generalize beyond the answer token?

The previous section shows that spline fitting is not necessary for the reported steering results. We now examine what those results establish by answering the following question:

Do these interventions manipulate a reusable concept representation?

We make the point that:

1. The causal evidence provided in the experiments is weak because the steering intervention is applied in a late layer (28) where the model's representation is already committed to a given token in each context.

2. The experiments provide very limited evidence of useful applicability of the steering intervention across various contexts. Tentative experiments to collect evidence of generalization across contexts do not provide very satisfactory results.

3.1 Late-layer steering resembles answer patching

The paper shows that manifold steering produces coherent intermediate output distributions. However, the intervention is applied at layer 28, close to the output of the model, and at the position used to predict the answer token. The result may therefore reflect an intervention on a late-stage answer representation rather than on a concept representation used across contexts.

To examine this possibility, we use the logit lens[4] [4] to measure when the correct answer becomes decodable from the residual stream. Although the logit lens is only an approximation to the model's intermediate computation, it shows that the correct answer is already strongly decodable at layer 28 (Figure 3).

Therefore, at layer 28 each centroid is already strongly associated with its answer token. Hence, the resulting steering behavior cannot be proven to be different from or stronger than a late-stage answer swap.

Important

More generally, it is not very convincing to study steering interventions where the concept steered towards is directly the next token. This delivers weak evidence because a simple token swapping operation would yield the same result. A more interesting result would be to

[4]We also provide in Appendix A.2 the same curves using the pretrained Jacobian lens provided by Neuronpedia.

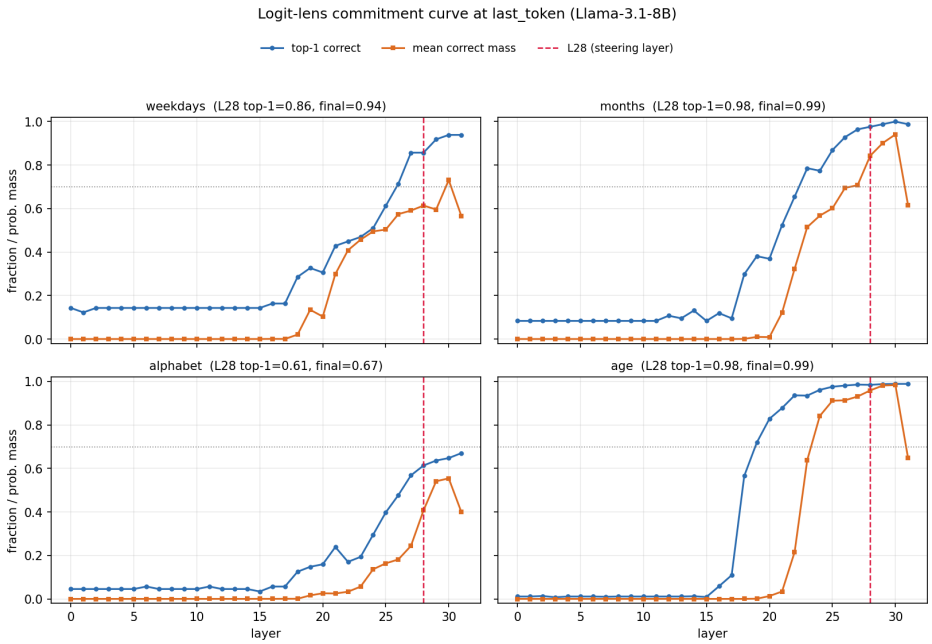


Figure 3: Commitment curves obtained using the logit lens.

measure the effect of steering on tokens different from the concept itself.

3.2 Disentangling concept value and output token

Setup. To separate the concept value from the output token, the model answers a two-option multiple-choice question by emitting `A` or `B`.[5] One option is correct and the other is a distractor. We steer from the correct value toward the distractor at the final "(" token, following [5]. We use the chat model and recompute centroids and steering paths from its activations on these prompts, with the correct answer balanced between `A` and `B` within each concept group. We report Δp, the increase in probability assigned to the distractor relative to the unsteered run.

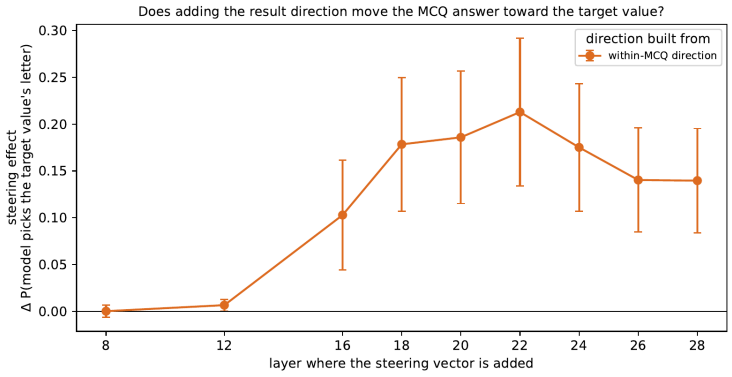


Figure 4: Δp as a function of the intervention layer.

[5]Prompt: `Q: What age is {number} years after {entity}? (A) {optA} (B) {optB}\nAnswer: (`

Results. Δp peaks at about 0.2 at intermediate layers (Figure 4). This is consistent with Section 3.1: at layer 28 the answer is already committed. The effect is far from the near-complete control obtained in the direct setting, although that comparison also differs in model and layer. It also grows with the distance between the two options, which equals the length of the steering path, and shrinks as the arithmetic offset increases (Figure 5).

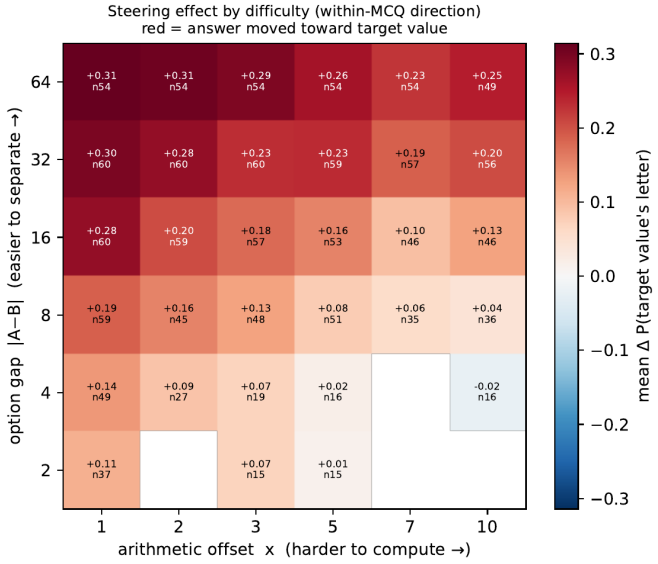


Figure 5: Δp as a function of the distance between options and the arithmetic offset.

Conclusion

The intervention transfers to a format where the concept value is not the output token, but it shifts at most about 20% of the probability mass, so it does not provide reliable control of the model's answer.

4 Conclusion

By reproducing and extending the experiments in [1], we obtain three main results:

- The reported all-pairs score does not establish a scaled isometry between activation and behavior spaces because it is dominated by the ordering used to fit both manifolds. A correspondence-permutation test nevertheless finds significant alignment between adjacent distances. The two spaces therefore share some local metric structure, although the current results do not establish distance preservation up to a common scale factor.
- Nonlinear manifold fitting is not necessary to reproduce the reported steering behavior. Piecewise-linear interpolation between neighboring centroids produces similar intermediate output distributions, both with and without the PCA projection. We obtain the same qualitative result on the two-dimensional grid task.
- The original experiments provide limited evidence that manifold steering acts on a context-independent concept representation. At the intervention layer, the answer is already strongly

decodable, and the target concept is also the next token. Our multiple-choice experiment provides partial evidence of generalization: the intervention retains an effect when the model outputs an option label rather than the concept value, but the effect depends strongly on the intervention layer, the distance between the answer options, and the arithmetic offset.

References

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[2] Joshua Engels et al. *Not All Language Model Features Are One-Dimensionally Linear*. arXiv:2405.14860 [cs.LG]. Feb. 2025. DOI: 10.48550/arXiv.2405.14860. URL: http://arxiv.org/abs/2405.14860 (visited on 06/10/2026).

[3] Dhruva Karkada et al. *Symmetry in language statistics shapes the geometry of model representations*. Version Number: 3. July 2026. DOI: 10.48550/ARXIV.2602.15029. URL: https://arxiv.org/abs/2602.15029 (visited on 07/04/2026).

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[5] Nina Panickssery et al. *Steering Llama 2 via Contrastive Activation Addition*. arXiv:2312.06681 [cs.CL]. July 2024. DOI: 10.48550/arXiv.2312.06681. URL: http://arxiv.org/abs/2312.06681 (visited on 07/15/2026).

[6] Wes Gurnee et al. *Verbalizable Representations Form a Global Workspace in Language Models*. arXiv:2607.15495 [cs.CL]. July 2026. DOI: 10.48550/arXiv.2607.15495. URL: http://arxiv.org/abs/2607.15495 (visited on 09/18/2026).

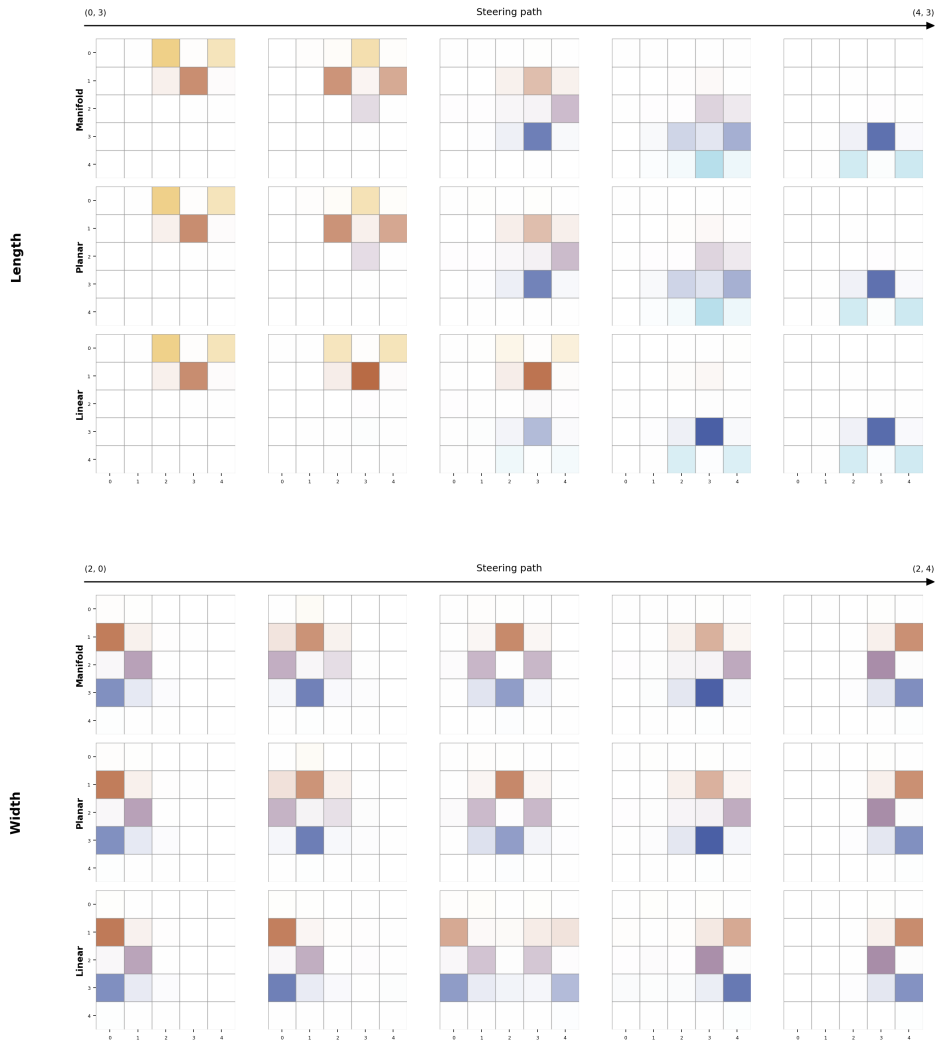


Figure 6: Comparison of steering methods on the two-dimensional grid task. Piecewise-planar interpolation reproduces the qualitative behavior of the fitted-manifold intervention.

A Additional experiments

A.1 Additional experiments on multi-dimensional concepts

Extending ablations to multi-dimensional concepts. The point of manifold steering was to go beyond linear steering. When steering between two grid values, we do not necessarily traverse intermediate values of the concept (as is the case in 1D), so in that case manifold fitting may be more relevant and reveal additional concept structure beyond intermediate values.

We reproduce the setting from Section 4 in [1] and refer to their paper for details about the experiment. As in the one-dimensional case, we compare the manifold fit to linear interpolation between neighboring centroids, which builds a piecewise planar structure in this 2D case.

We observe that Figures 6–7 confirm and extend the one-dimensional ablation. Indeed, the additional PCA and spline machinery does not yield a visible advantage over local linear interpolation on the two-dimensional task.

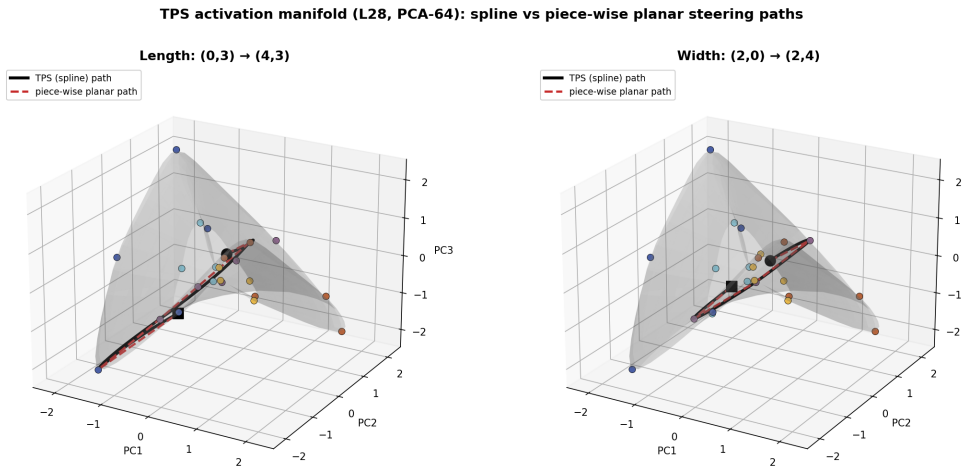


Figure 7: Steering paths through the grid representation. The fitted manifold and the piecewise-planar construction traverse similar regions of activation space.

A.2 Experiment with the Jacobian lens

We reproduce the logit lens experiment with the Jacobian lens [6] for completeness. The results obtained are very similar and confirm the conclusions from Figure 3[6].

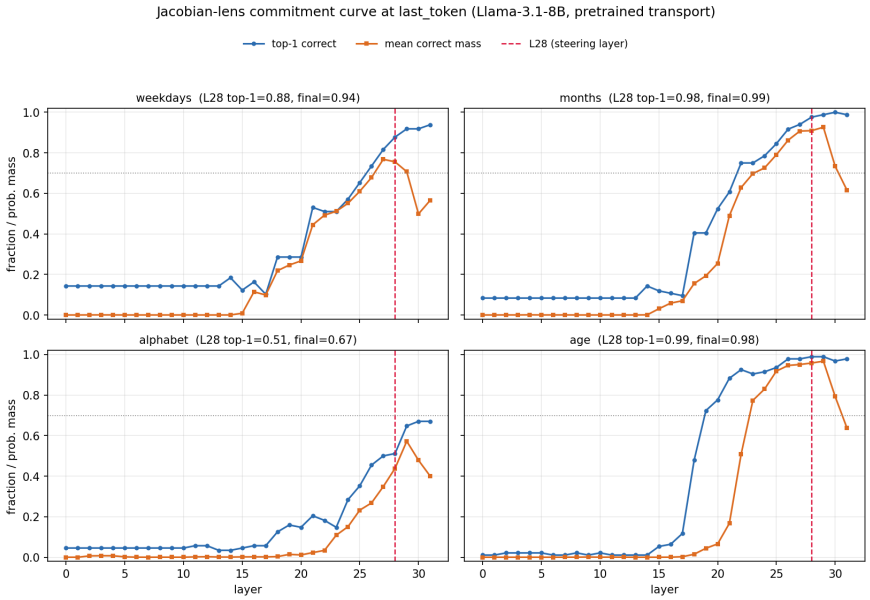


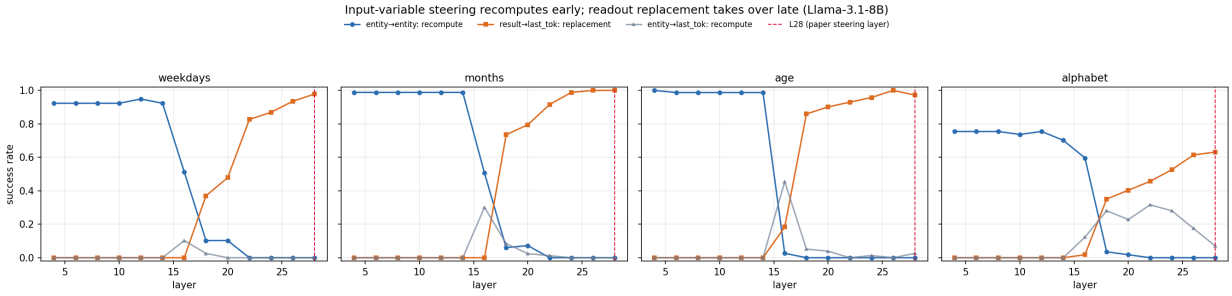
Figure 8: Commitment curves obtained using the Jacobian lens.

[6] An interesting (unrelated) observation is the sudden drop of the mean correct mass on the last layer. One plausible explanation is that the last layer mainly performs logit calibration, and that the probabilities at earlier layers overestimate the final result because they come before logit calibration. The Jacobian lens does not mitigate this effect, as it specifically excludes the last layer (or last two) for this reason.

A.3 Manipulating inputs vs. outputs

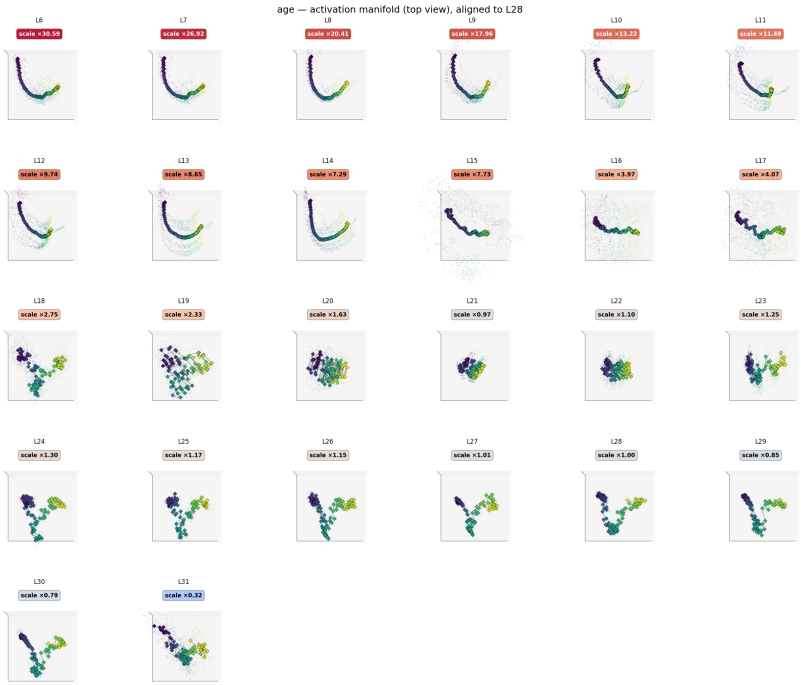
A simple alternative to the experiments presented in the paper consists in steering the input of a given prompt instead of steering the expected answer towards another answer. For instance, for the prompt "What is three days after Monday?", the paper applies the steering intervention Thursday $\rightarrow$ Friday and measures whether the answer obtained is indeed Friday. We perform the symmetric experiment by steering the same prompt using the intervention Monday $\rightarrow$ Tuesday and expect the answer to become Friday. Of course, the steering intervention is now performed at the Monday token, and is expected to work at early layers.

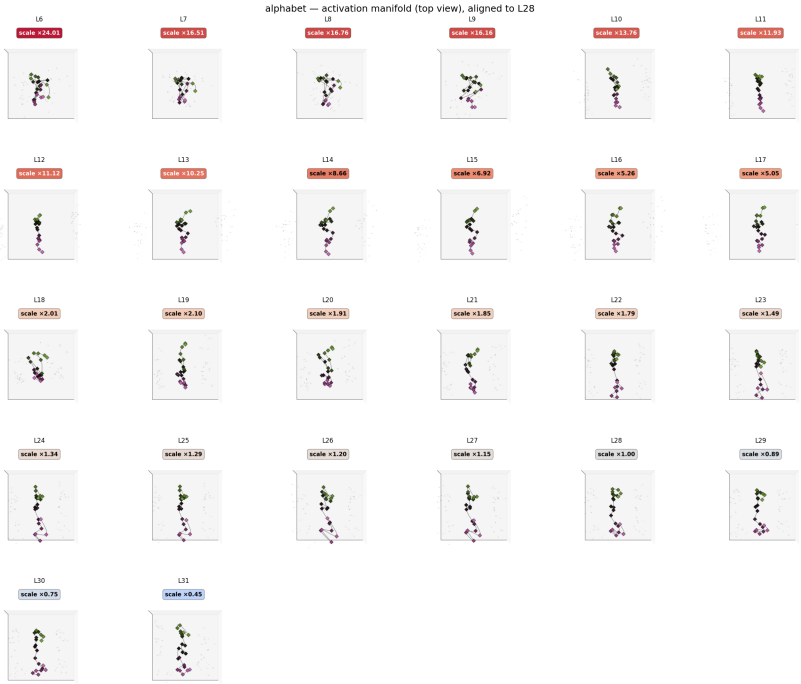
We observe the complementary behavior of these two related interventions in the plot below:

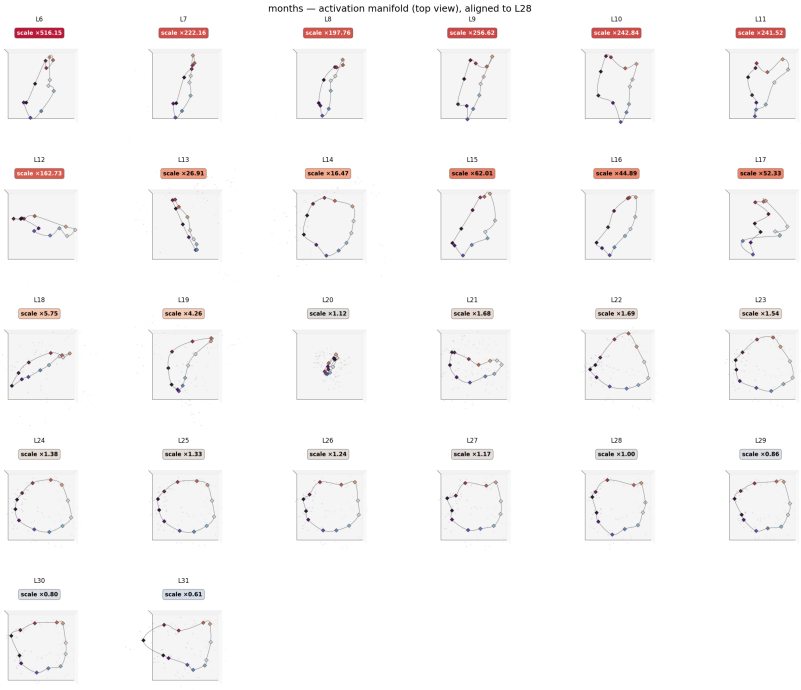


A.4 Visualizing the manifolds across layers

We plot for each concept the manifold built at each layer. We rescale the manifold to share the same scale as the one at layer 28 and report the scaling factor applied.




alphabet — activation manifold (top view), aligned to L28
L6
scale ×24.01
L7
scale ×16.51
L8
scale ×16.76
L9
scale ×16.16
L10
scale ×13.76
L11
scale ×11.93
L12
scale ×11.12
L13
scale ×10.25
L14
scale ×8.66
L15
scale ×6.92
L16
scale ×5.26
L17
scale ×5.05
L18
scale ×2.01
L19
scale ×2.10
L20
scale ×1.91
L21
scale ×1.85
L22
scale ×1.79
L23
scale ×1.49
L24
scale ×1.34
L25
scale ×1.29
L26
scale ×1.20
L27
scale ×1.15
L28
scale ×1.00
L29
scale ×0.89
L30
scale ×0.75
L31
scale ×0.45


months — activation manifold (top view), aligned to L28
L6
scale ×516.15
L7
scale ×222.16
L8
scale ×197.76
L9
scale ×256.62
L10
scale ×242.84
L11
scale ×241.52
L12
scale ×162.73
L13
scale ×26.91
L14
scale ×16.47
L15
scale ×62.01
L16
scale ×44.89
L17
scale ×52.33
L18
scale ×5.75
L19
scale ×4.26
L20
scale ×1.12
L21
scale ×1.68
L22
scale ×1.69
L23
scale ×1.54
L24
scale ×1.38
L25
scale ×1.33
L26
scale ×1.24
L27
scale ×1.17
L28
scale ×1.00
L29
scale ×0.86
L30
scale ×0.80
L31
scale ×0.61


weekdays — activation manifold (top view), aligned to L28
L6
scale ×784.21
L7
scale ×564.12
L8
scale ×508.52
L9
scale ×380.95
L10
scale ×328.29
L11
scale ×350.35
L12
scale ×383.98
L13
scale ×194.41
L14
scale ×122.75
L15
scale ×83.32
L16
scale ×78.48
L17
scale ×44.02
L18
scale ×7.71
L19
scale ×4.37
L20
scale ×2.37
L21
scale ×1.63
L22
scale ×1.44
L23
scale ×1.35
L24
scale ×1.28
L25
scale ×1.22
L26
scale ×1.18
L27
scale ×1.13
L28
scale ×1.00
L29
scale ×0.89
L30
scale ×0.80
L31
scale ×0.64